

Vu Bao Khanh

HCMC, Vietnam | (+84)-0947546007 | vubaokhanh2311@gmail.com | <https://github.com/vubaokhannh>

Fresher Node.js Backend Developer with hands-on experience building scalable RESTful APIs using NestJS, PostgreSQL, and Prisma. Experienced in implementing JWT authentication, RBAC systems, and real-time collaboration features with Socket.IO. Passionate about clean architecture, performance optimization, and eager to contribute to production-grade backend systems.

WORK EXPERIENCES

Frontend Developer – CUSC (Cantho University Software Center) May 2025 – August 2025

- Tech stack: Next.js, Mantine UI, Tailwind CSS, Axios
- Built responsive and user-friendly frontend interfaces ensuring smooth user experiences across devices.
- Integrated RESTful APIs with Axios to fetch and display dynamic data in real time.
- Collaborated with backend developers to optimize data flow and improve system performance.
- Implemented state management and reusable UI components to enhance maintainability and scalability.

PERSONAL PROJECTS

2025 – Present: Task Management System

Backend (NestJS)

- **Tech stack:** NestJS, TypeScript, PostgreSQL, Prisma, JWT, Socket.IO, Redis, Firebase Cloud Messaging
- Designed and developed RESTful APIs for Boards, Lists, Cards, Labels, Members, Comments, and Attachments using NestJS.
- Implemented secure JWT-based authentication and authorization flows including register, login, refresh token, forgot password (via email), reset password, and logout.
- Built scalable board-level access control (Owner / Editor / Viewer) and system-level RBAC using Role, Permission, and RolePermission models.
- Designed relational database schema with PostgreSQL and Prisma ORM, ensuring data consistency and efficient querying.
- Developed real-time collaboration features using Socket.IO for board updates, lists, cards, labels, members, comments, and attachments.
- Integrated Redis caching to reduce database load and improve API response performance.
- Implemented push notifications using Firebase Cloud Messaging (FCM) for offline users.
- Applied NestJS modular architecture to improve code maintainability and scalability.
- Optimized Prisma queries to prevent N+1 query issues and improve performance.
- Documented APIs using Swagger (OpenAPI).
- Deployed backend on an Ubuntu VM (Proxmox) and configured Cloudflare DNS to expose public RESTful APIs.
- Implemented CI/CD pipeline using GitHub Actions (self-hosted runner) to automate build and deployment via SSH and PM2.
- **GitHub:** <https://github.com/vubaokhannh/trello-backend>
- **Link demo:** <https://api.krello.biz/api-docs#/>

Frontend (React)

- **Tech stack:** React, TypeScript, Axios, Mantine UI, Tailwind CSS, Zustand, Socket.IO
- Built drag-and-drop board interface using @dnd-kit for managing boards, lists, and cards.
- Integrated backend RESTful APIs using Axios for authentication and real-time data synchronization.
- Managed global state using Zustand for better scalability and performance.
- Developed responsive and modern UI using Mantine UI and Tailwind CSS.
- Implemented real-time UI updates for collaborative scenarios.
- **GitHub:** <https://github.com/vubaokhannh/trello-frontend>
- **Link demo:** <https://web.krello.biz/>

TECHNICAL SKILLS

- **Backend:** Node.js, NestJS, TypeScript, Prisma
- **Frontend:** React, Tailwind CSS, Mantine UI
- **Database:** PostgreSQL, MySQL, Redis
- **Realtime & Messaging:** Socket.IO, Firebase Cloud Messaging
- **Concepts:** RESTful API, MVC Architecture, OOP, Authentication & Authorization, RBAC
- **Tools:** Git, GitHub, Postman, Docker, Figma

EDUCATIONAL BACKGROUND

FPT Polytechnic Can Tho – Major: Web Developer (Aug 2023 – Aug 2025)

GPA: 3.75/4.0